



AI & DATA SCIENCE USING PYTHON

Introduction to AI ML DS , Gen AI and Agentic AI (DEMO)

- Relationship between AI, ML, DL, NLP, Data Science, Gen AI, Agentic AI
- Applications for data science
- Why Python for Data Science
- AI Tools: Claude/Perplexity/Co Pilot/ChatGPT/Deep Seek

Python Programming

Python Overview

- Installation of Anaconda Python,
- Python Features
- Variables, Operators, Data Types
- Conditions
- Loops
 - Functions
 - Modules, Packages
- String object, Exercises
 - List object, Exercises, Case study
- Tuple object, Exercises
- Dictionary object, Exercises, Case Study
- Set, Frozenset
- Comprehensions
 - List comprehension,
 - Dictionary comprehension.
 - Set Comprehension
- Python File I/O
 - Applications using File IO
 - Exercises

- Regular expressions
 - Identifiers
 - Quantifiers
 - Lookahead, Lookbehind examples
 - Exercises
- SQL Database SQLITE
 - CURD operations, SQL Queries, Kinds of Joins
 - **Project on SQL database with Sqlite**
- SQL Database MySQL
 - Environment setup, CURD operations, SQL Queries, Kinds of Joins
 - **Project on SQL database with Sqlite**
- Working with JSON data
- Working with XML data,
- Working with PDF data
- Working with CSV data
- Generators, Iterators,
 - Generator functions and Generator Expressions
 - Comparison between generator expression and List comprehensions
- Collections Module
 - Counter, OrderedDict, Defaultdict, Chainmap, NamedTuple
 - Hacker Rank problem solving,
 - Codechef problem solving
- **Object Oriented Python,**
 - Class, Object
 - Abstraction
 - Encapsulation
 - Inheritance
 - Polymorphism
 - Case studies

Mathematics for Data Science

Mathematical Computing with Python (NumPy)

- Introduction to NumPy, N-D array
- Data types and attributes of Arrays
- Mathematical Functions of NumPy
- Array Indexing and Slicing
- Array broadcasting
 - Comparing Core Python Objects with Numpy
 - Exercises
 - Case study

Statistics

- Central Tendency (mean, median and mode)
- Measures of Variation (Interquartile Range, Variance, Standard Deviation)
- Bar Chart, Histogram, Box whisker plot, Scatter Plot
- Co-variance, Correlation
- Central Limit Theorem,
- Skewness and Kurtosis
- Z Test, T Test, P-value,
- Hypothesis testing
- Chi-Square, F- Statistic
- Sampling Techniques
- ANOVA (Analysis of Variance)

Probability

- Introduction to Probability, Uncertainty, Random numbers
- Joint Probability, Marginal Probability, Conditional Probability, Exclusivity
- Probability Distributions (PMF, CDF, Normal Distribution, ..)
- Bayes Theorem.

DATA ANALYTICS

- **Introduction to Pandas**

- Understanding DataFrame
- Transforming List, Tuple, Dictionaries into Data Frame
- Accessing rows and columns, Iteration over Data Frames
- Pandas joining and merging,
- Pandas Groupby, Pivot Table, Binning,
- Pandas Visualization
- Data Generation
 - **Real time Case Studies on Data Analysis based on Kaggle**

Exploratory Data Analysis and Data Visualization

- Univariant analysis
 - Numerical, Categorical
- Bivariant analysis
 - Numerical-Numerical
 - Numerical-Categorical
 - Categorical-Categorical
- Data Visualization
 - Matplotlib data visualization
 - Seaborn data visualization

Power BI

- Business Intelligence (BI)
- Loading data into PowerBI
- Working with Power Query Editor
- Working with Report Section & Visuals in Power BI

Front end and Web Application development

User Interface for Model Deployment

- HTML:- Basic Elements, Lists, Tables, Forms, Examples
- CSS: Syntax, Selectors, inline/internal/external CSS examples,
- Flask Framework: Environment,
Routing, URL building, HTTP methods, Templates,
Static files, Request object, Flask with Sqlite database, case study
- Streamlit Framework, Gradio Framework
- FastAPI
 - Pydantic for data validation
- Web scraping using beautifulsoup4, requests libraries
 - Case Studies on web scraping.

Machine Learning

Data Preprocessing Techniques

- Data Imputation (Missing values)
 - Simple Imputation
 - KNN Imputation
 - Iterative Imputation
- Data Encoding Techniques
 - Label Encoding
 - OneHot Encoding
- Finding Outliers
 - IQR technique
 - Zscore technique
 - Percentile technique
 - LocalOutlierFactor

- Data Normalization, Transformation, Scaling
 - MinMaxScaler
 - StandardScaler
 - RobustScaler
 - PowerTransform
 - Box-CoxTransform
 - QuantileTransforms
- Dimensionality Reduction Techniques
 - PCA- Principle Component Analysis
 - SVD- Singular Value Decomposition
 - LDA- Linear Discriminant analysis
- Feature Selection (Importance) and Engendering techniques
 - Supervised learning based
 - Unsupervised learning based
- **Case Studies on Data Preprocessing techniques and comparative analysis of various techniques**

Regression Analysis

- Regression, Linear regression
- Linear regression, Multiple Regression
- Ridge Regression, Lasso Regression
- Explanation of statistics
- Evaluation metrics (R-Square, Adj R-Square, MSE, RMSE)
- Train/Test Split, Hypothesis testing formal way
- Case Studies
- **Project on Regression Analysis from Kaggle**

Classification

- Introduction to Machine Learning
- Naïve Bayes classifier
- Decision Tree classifier

- KNN classifier
- Logistic Regression
- Support Vector Machines (SVM)
- One-vs-Rest and One-vs-One for Multi-Class Classification
- Predict() Vs PredictProba()
- Ensemble models (Random Forest, Bagging, Boosting)
 - Bagging algorithms
 - Boosting algorithms
 - Stacking algorithms
 Xgboost indepth with Industry cases
- SK Learn ML library using Python and Case Studies
- **Project on Classification Algorithms from Kaggle**

Model Selection and Evaluation

- Accuracy measurements
- Precision, Recall, Precision – Recall Tread-off
- AUC Score, ROC Curve
- Train/Validation/Test split, K-Fold Cross Validation
- The Problem of Over-fitting (Bias-Variance tread-off)

Learning Best Practices for Model Evaluation

- Bias, Variance, Overfitting, Underfitting methods
- Pipelining
- Parameter Tuning mechanisms (Grid Search, Random Search)
- Debugging algorithms with learning and validation curves

Case Study

Association Analysis

- Association Rules & Interesting measures
- Apriori Algorithm
- FP-Growth algorithm
- Case Studies

Clustering

- Similarity distance measures
- K-means Clustering
- Hierarchical Clustering
- DB Scan Clustering
- Case Studies

MLOPS

- MLFLOW , End to End ML case Studies
- DVC

Applications of ML

- Time Series Analysis (Stock Market forecasting using ARIMA models)
- Recommendation Systems (Filter based RS and Collaborative based RS)
- Dealing with Imbalanced datasets (Anomaly Detection Methods)

Deployment ML/DL/NLP models using Flask, Fast API, Github

ML Model Deployment in Azure

- Introduction to Cloud Computing and Azure
- Azure Machine Learning Studio Basics
- Model Deployment via Notebooks, Designer, Automated ML

Natural Language Processing

NLP

- NLP Overview, Applications using NLTK, Text Blob
- Tokenizing, Stop Word Removal, Stemming, Lemmatization, POS Tagging,
- Similarity measures over Text,
- Vector Space Model, Bag of words,
 - transforming text to Numeric using Count Vectorizer
- Text Classification,
 - Text Clustering,
 - Topic Modelling,
 - Model Deployment using NLP
- Word Embeddings, Sentiment Analysis
 - Case Studies
 - **Project on NLP from Kaggle**

Computer Vision

Image Processing using CV2

- Image Processing Basics and Computer Vision Library
- Images Operations using Numpy
- Edge detection
- Contour detection
- Feature Marching
- Face detection

DEEP LEARNING

Deep Learning using TensorFlow and Kera's.

- Introduction to Deep Learning, Neurons, Perceptron, Multilayer Perceptron,
- Forward Propagation, Backward Propagation, Activation Functions,
- Artificial Neural Networks (ANN)
 - Case Studies
- Convolution Neural Networks (CNN)
 - Case Studies
- Recurrence Neural Networks (RNN)
 - LSTM, GRU algorithms
 - Case Studies
- Model Deployment using Deep Learning
- Deep Learning with Text
- Word Embedding
- Transformers
 - Encoders and Decoders,
 - Attention mechanism
- **Hugging Face Pre-trained models**

GEN AI , RAG Vector Databases, AGENTIC AI

Generative AI and LLMs with Use Cases

- Gen AI text, Images, Video
- Creating own Chatbots
- Gen AI using SQL function calling
- Gen AI with Word Embeddings and case studies
- **Gemini, Llama, Ollama, gemma , Mistral etc**
- Vector Databases
 - Chroma DB
 - Faiss,
 - Pinecone
- Retrieval-Augmented Generation (RAG)
 - Applications of RAG
- **Conversation between Llama/Gemini with Notepad text, PDF, Web data.**

Agentic AI Frameworks

- AI Agents, Agentic AI
- Introduction to Langchain
 - Basic Components and Modules
 - Data Ingestion & Processing
 - Data Transformers
 - Embeddings & Vector Stores with Langchain
 - Creating chatbots and Agents
- Build expressive, customizable agent workflows using LangGraph, Agentic RAG
- Introduction to LangSmith
- Introduction to CrewAI - Multi-Agent Platform
- Tools:- Autogen / n8n